



CLOUD COMPUTING LAB

Lab Number 01

**Lab Title: GitHub Identity, Student Portal Registration, and
Ubuntu Server Installation**

Name: Ayesha Asif

Registration No: FA24B1-SE-046

Course Code: SE-665

Section: B

GitHub Username: ayeshaasif46

DEPARTMENT OF SOFTWARE ENGINEERING

SESSION: 2024-2028

LAB TASKS

Task1: Create and Configure Your GitHub Account.

Description: Created a personal GitHub account using actual identity, set the username to ayeshaasif46, and updated the Public Profile settings to display the full name **Ayesha Asif**.

Figure1.1: github_profile.png

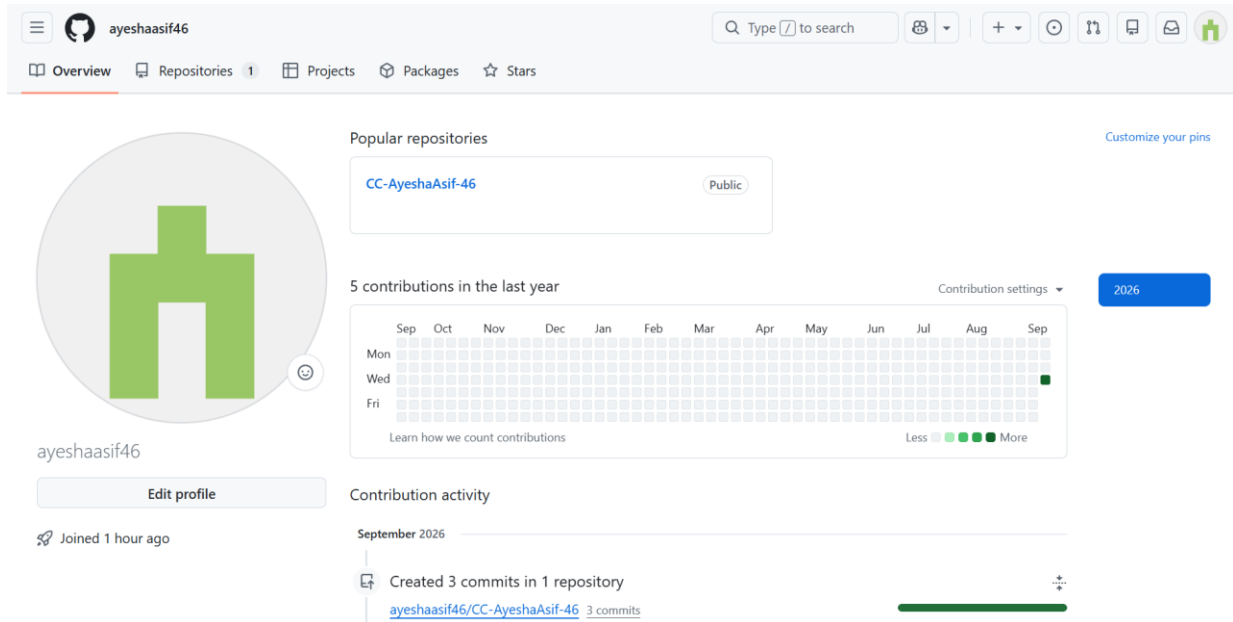
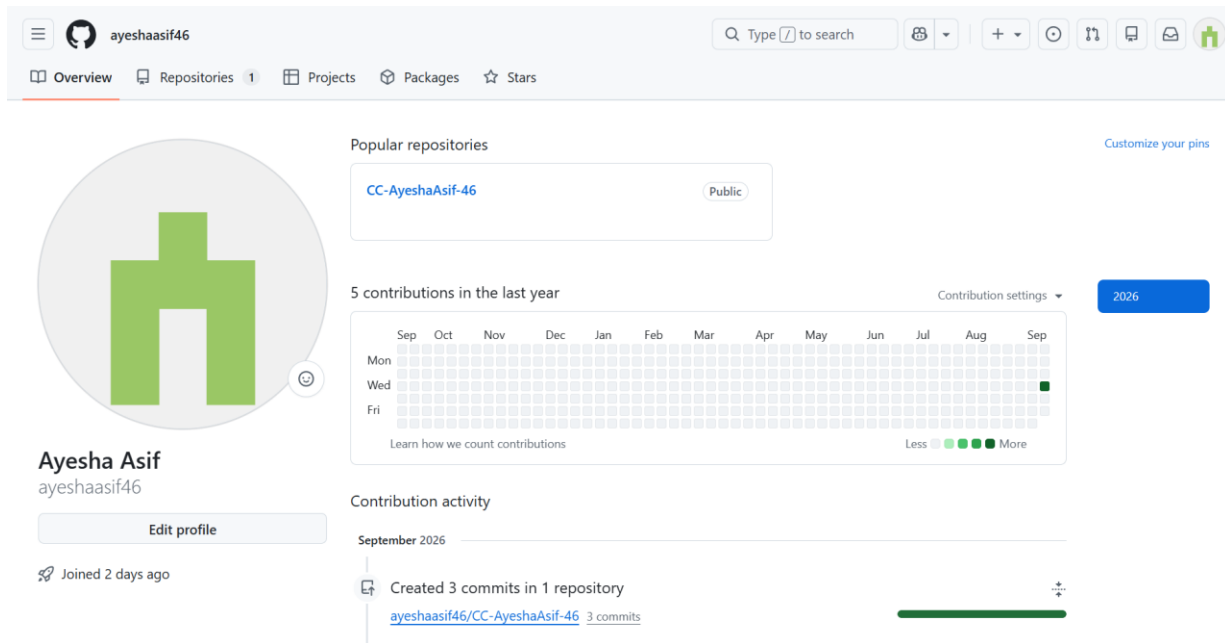


Figure 1.2: github_actual_name.png



Task 2: Register in the Student Marks Portal.

Description: Logged into the Student Marks Portal using the ayeshaasif46 GitHub account and submitted official registration details using Registration no 2024-BSE-46.

Figure 2.1: portal_login_page.png

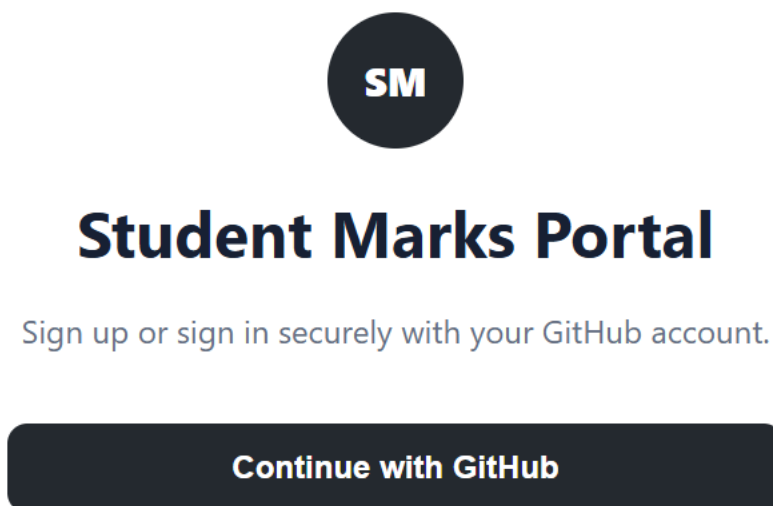


Figure 2.2:portal_github_profile.png



Complete registration

Enter your registration number, course and section.

The enrollment requires approval.

Registration number

 ▼

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

 ▼

Section

 ▼

Figure 2.3: portal_enrollment_submitted.png

Course enrollments

Sign out



Ayesha Asif

2024-BSE-46

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section B

Waiting for administrator approval

Pending approval

Task 3: Download VMware Workstation Pro and Ubuntu Server

Description: Installed VMware Workstation Pro, downloaded the Ubuntu Server, and verified over 20 GB of free disk storage space.

Figure 3.1: vmware_workstation.png

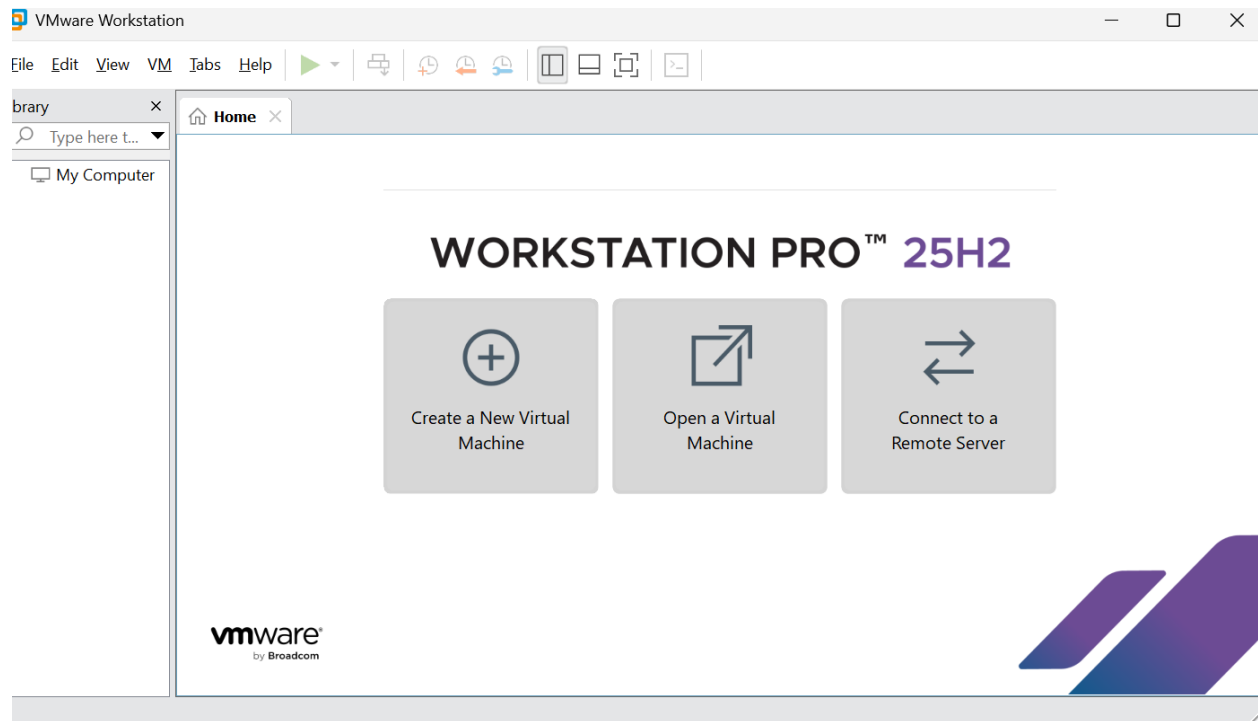
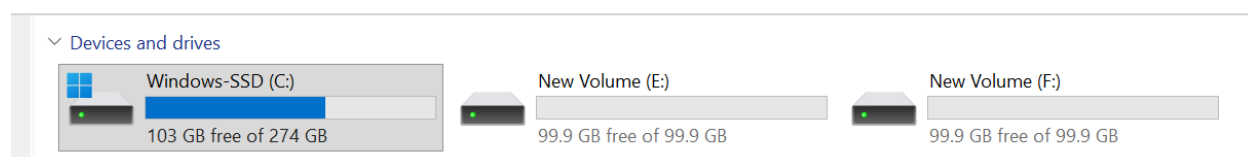


Figure 3.2: ubuntu_server_iso.png

 VMware-Workstation-Full-25H2u1-25219725	9/16/2026 1:37 PM	Application	284,992 KB
 ubuntu-26.04.1-live-server-amd64	9/16/2026 11:46 AM	Disc Image File	2,859,240 KB

Figure 3.3: available_storage.png



Task 4: Create the Ubuntu Server Virtual Machine

Description: Configured a new "Typical" virtual machine in VMware Workstation, selected the downloaded Ubuntu ISO file, allocated 20 GB of virtual disk space, and started the boot process.

Figure 4.1: vm_creation_wizard.png

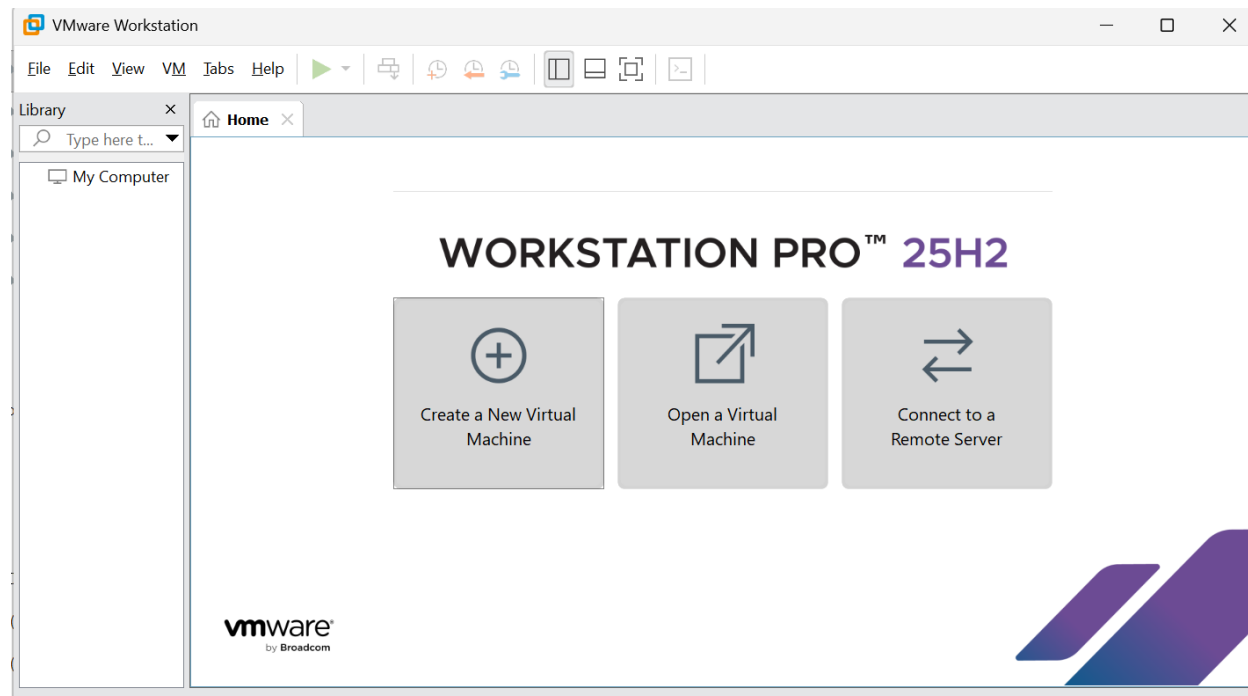


Figure 4.2: vm_typical_configuration.png



Figure 4.3: vm_ubuntu_iso_selected.png

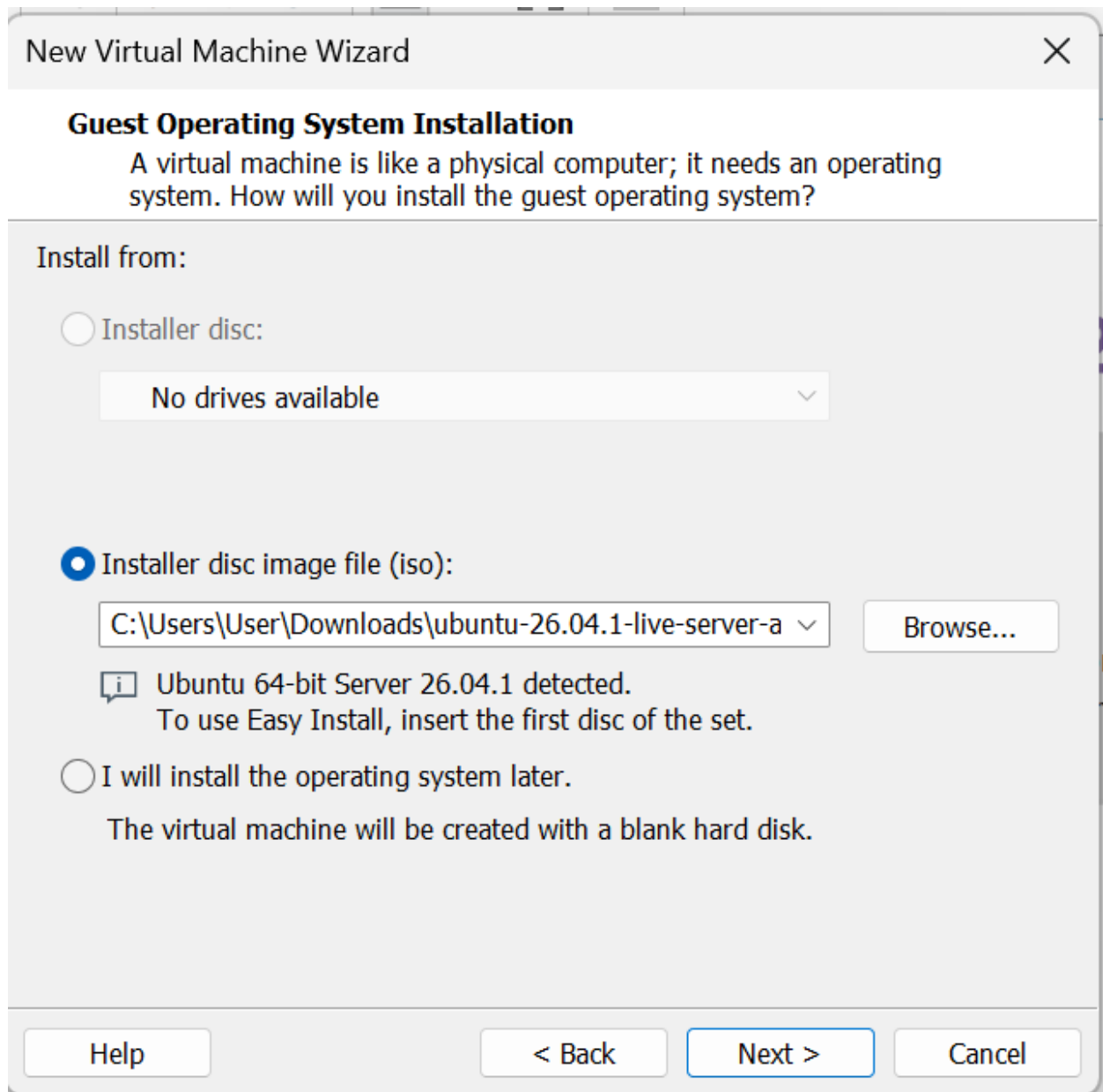


Figure 4.4: vm_configuration_summary.png

New Virtual Machine Wizard

×

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

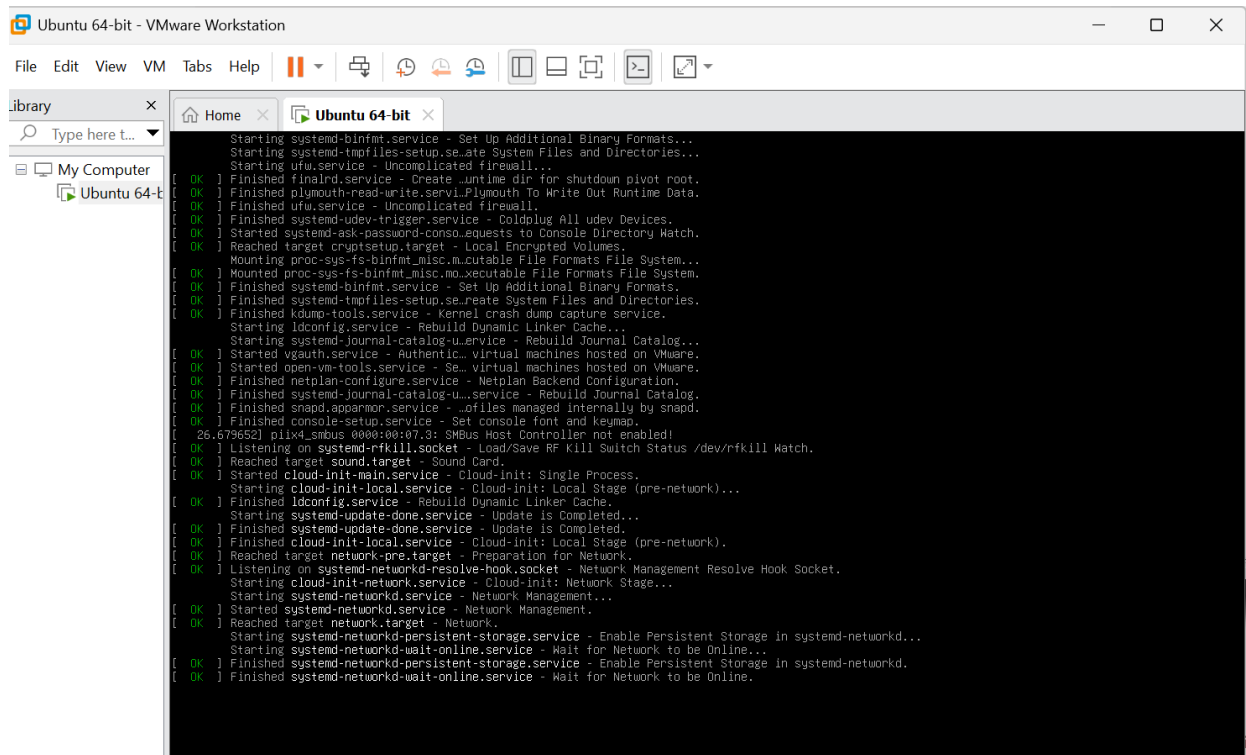
Help

< Back

Next >

Cancel

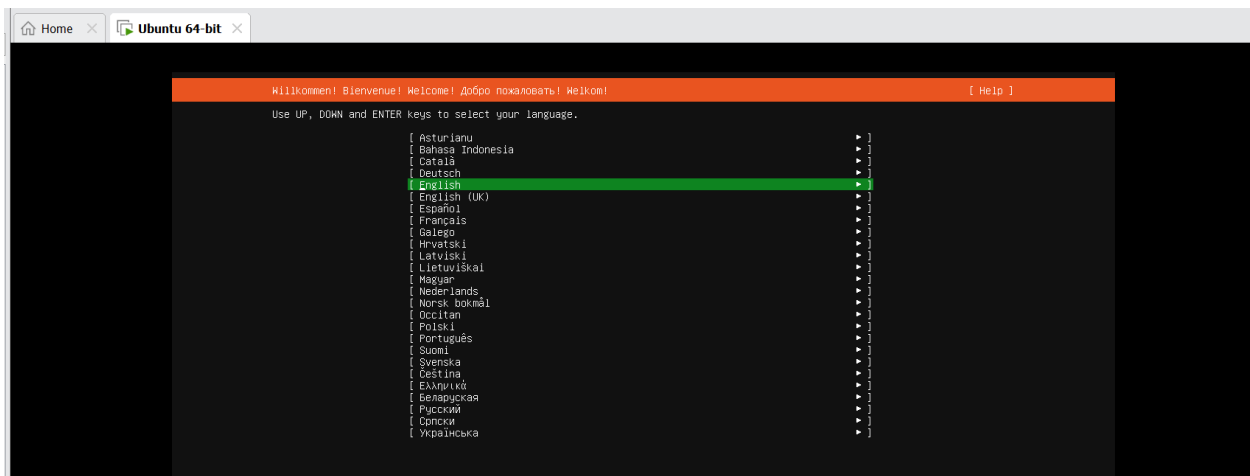
Figure 4.5: Ubuntu_installer_boot.png



Task 5: Choose the Installation Language

Description: Navigated the Ubuntu Server installer startup screen using arrow keys and selected English as the installation language.

Figure 5.1: Ubuntu_language.png



Task 6: Configure the Keyboard Layout

Description: Selected the English (UK) keyboard layout and variant to ensure accurate terminal typing.

Figure 6.1: ubuntu_keyboard_layout.png

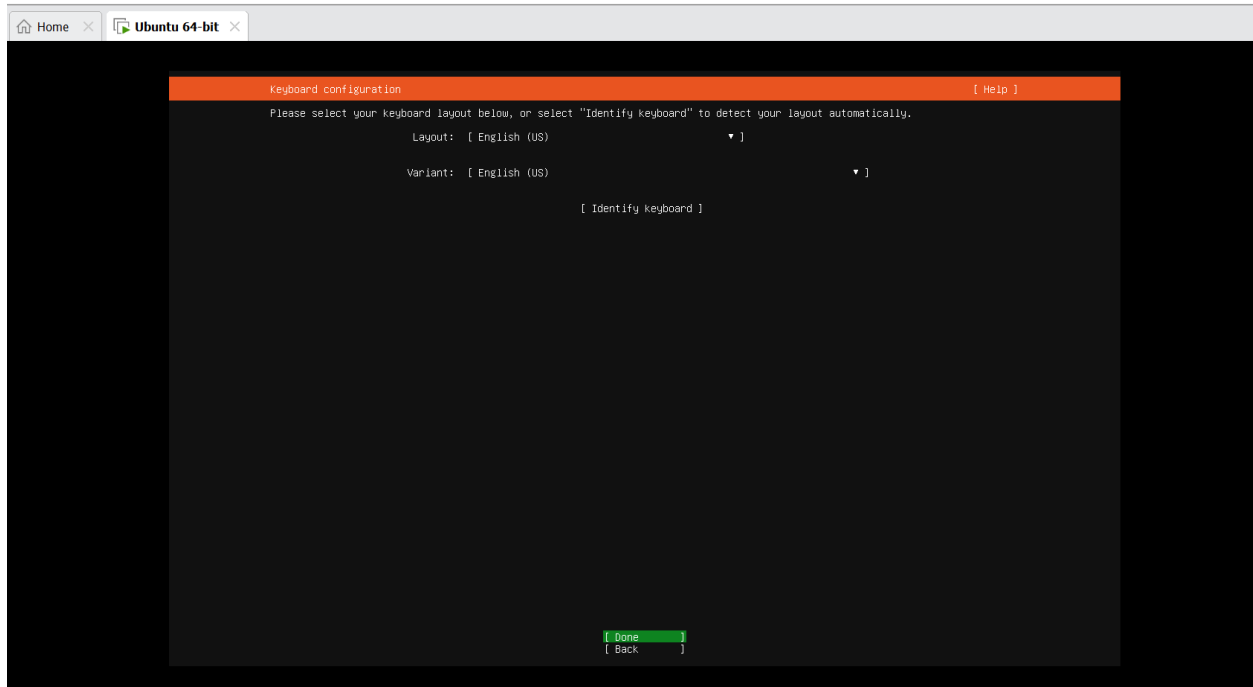
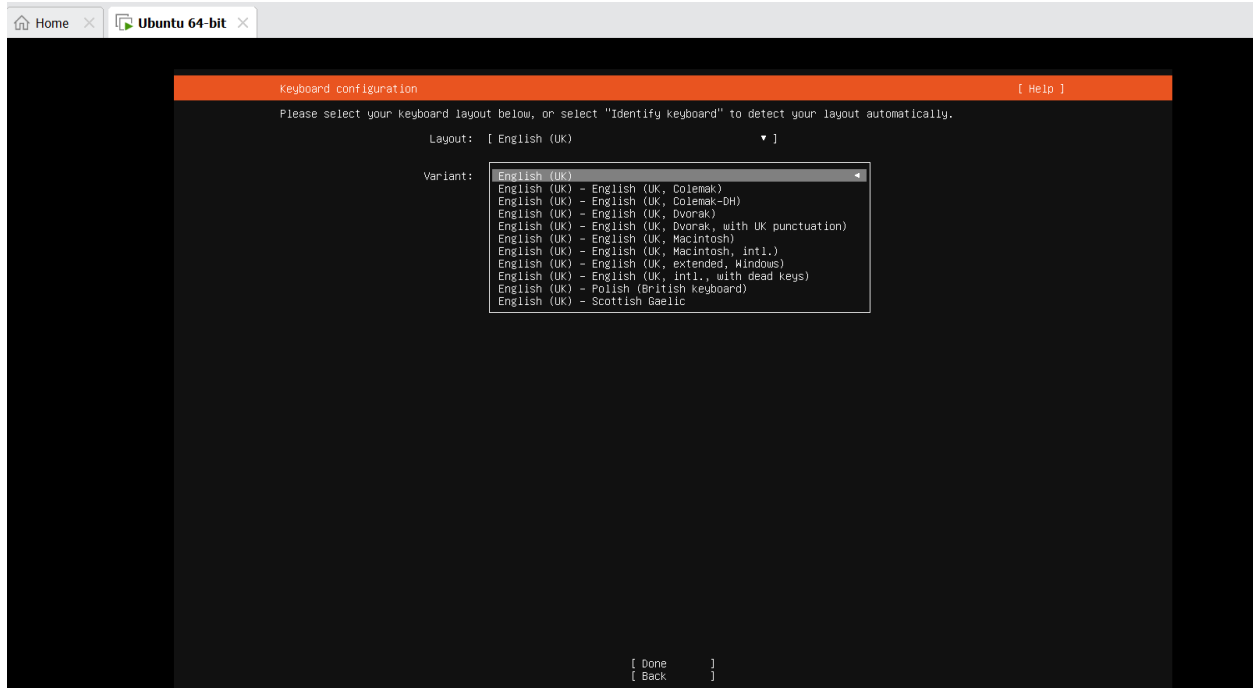


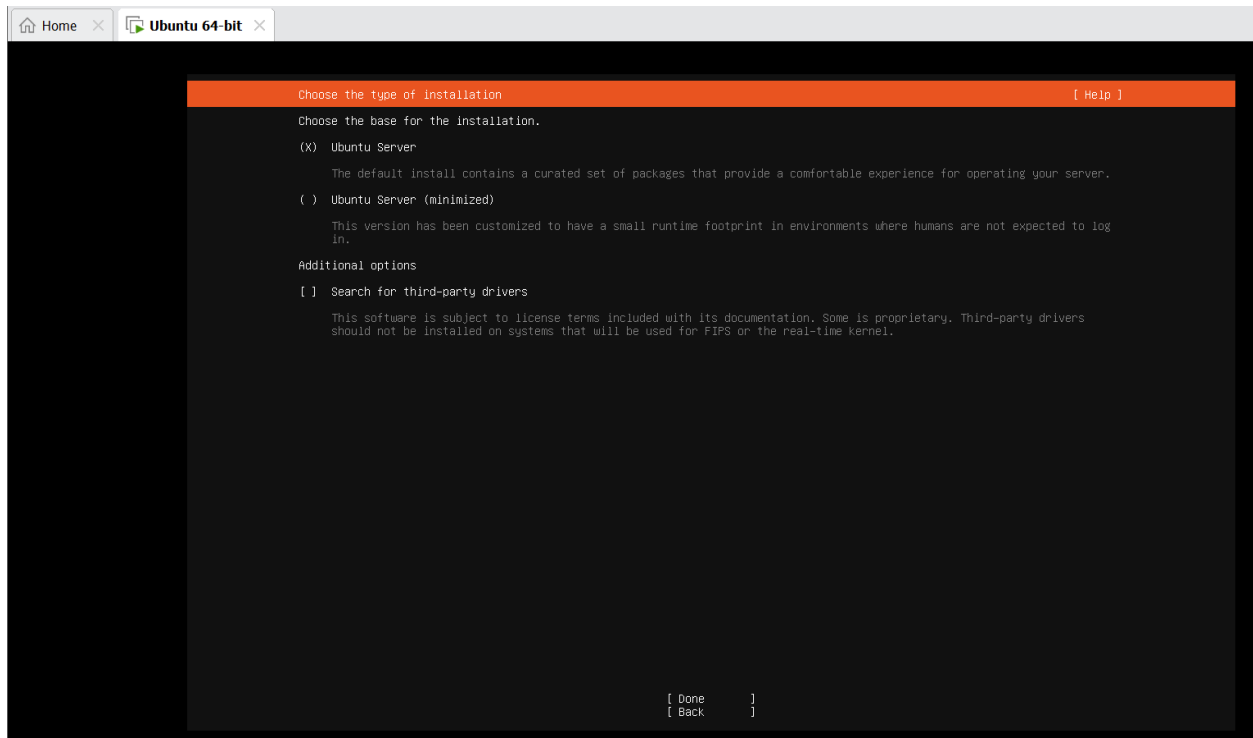
Figure 6.1: ubuntu_keyboard_variant.png



Task 7: Choose the Ubuntu Server Installation Type

Description: Selected the standard Ubuntu Server base installation profile, skipping unnecessary configurations.

Figure 7.1: ubuntu_installation_type.png



Task 8: Configure the Network

Description: Verified that the network interface was properly detected and received an automated IPv4 address via DHCP.

Figure 8.1:ubuntu_network_interface.png

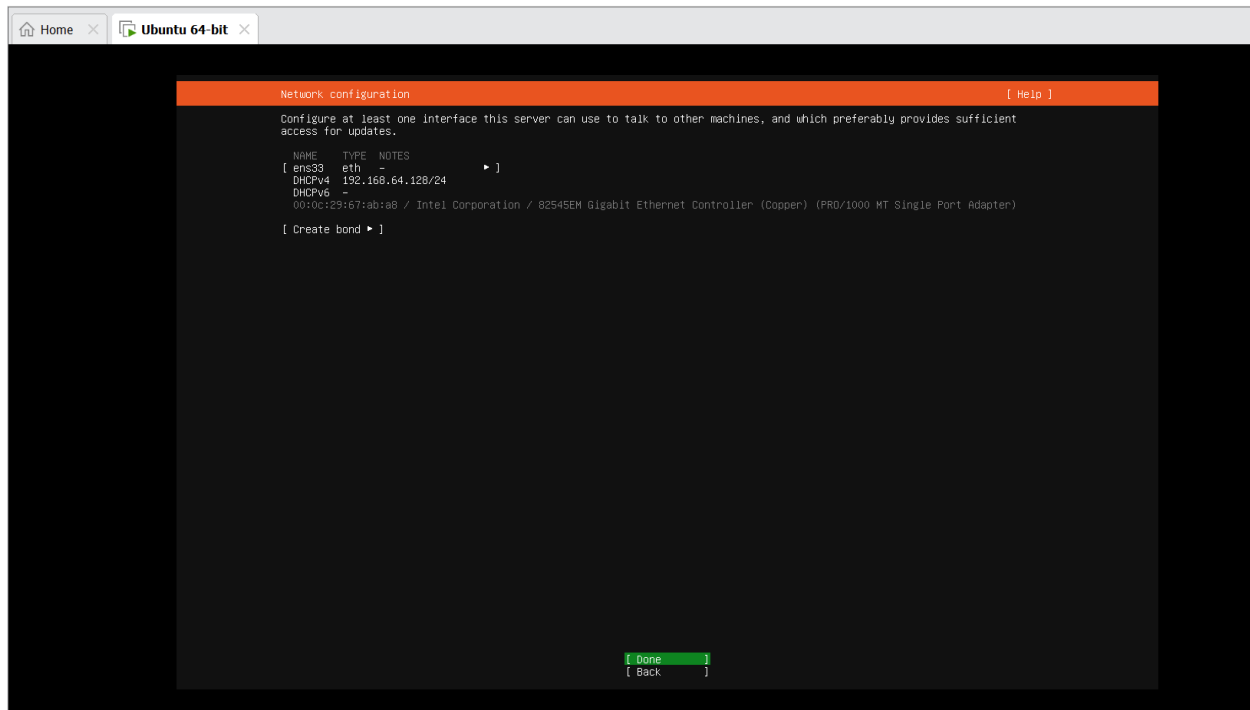
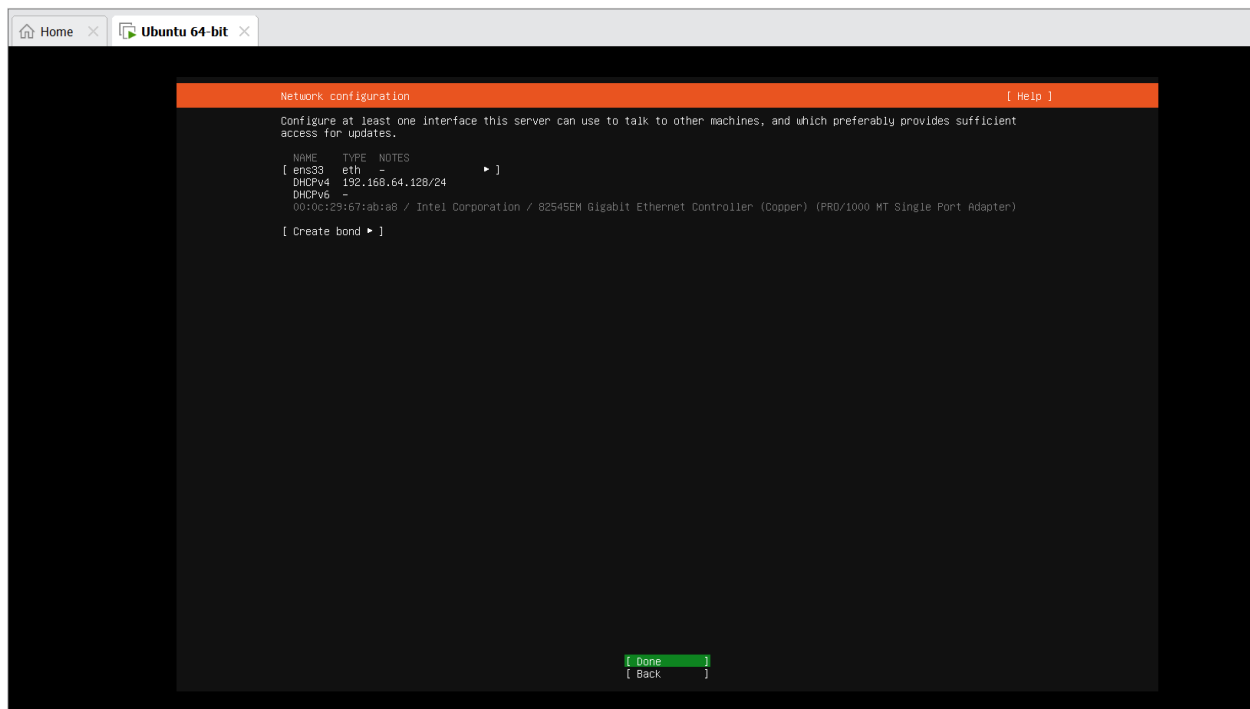


Figure 8.2:Ubuntu_installer_network.png



Task 9: Configure Storage

Description: Selected the guided storage option to allow automatic layout partitioning on the allocated virtual disk.

Figure 9.1: ubuntu_guided_storage.png

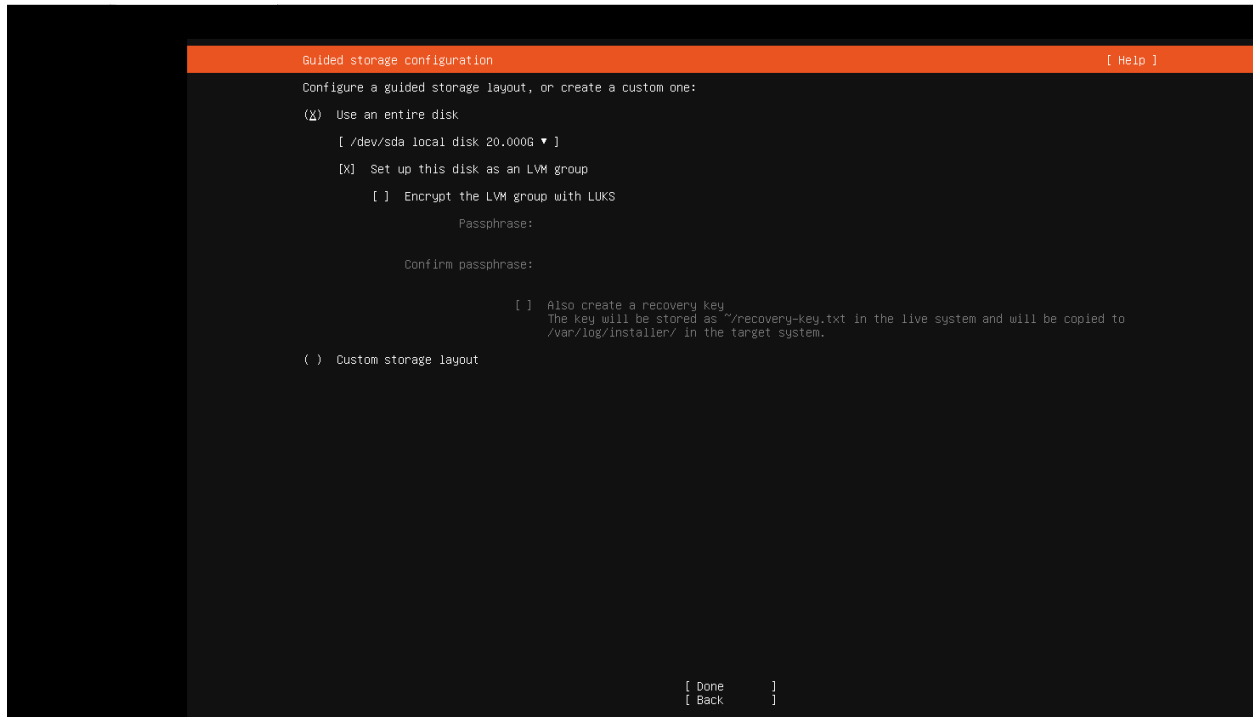
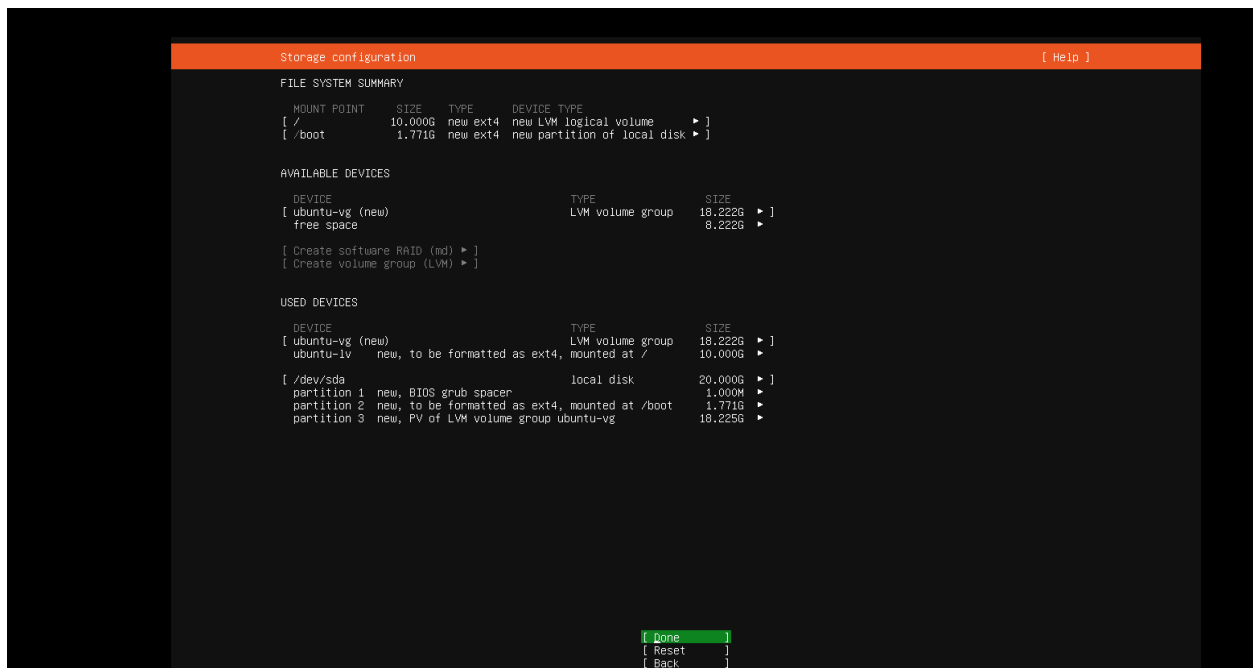


Figure 9.2: ubuntu_storage_configuraion.png



Task 10: Select the Installation Disk

Description: Selected the targeted 20 GB VMware virtual disk for installation after confirming its details and capacity.

Figure 10.1: ubuntu_selected_disk.png

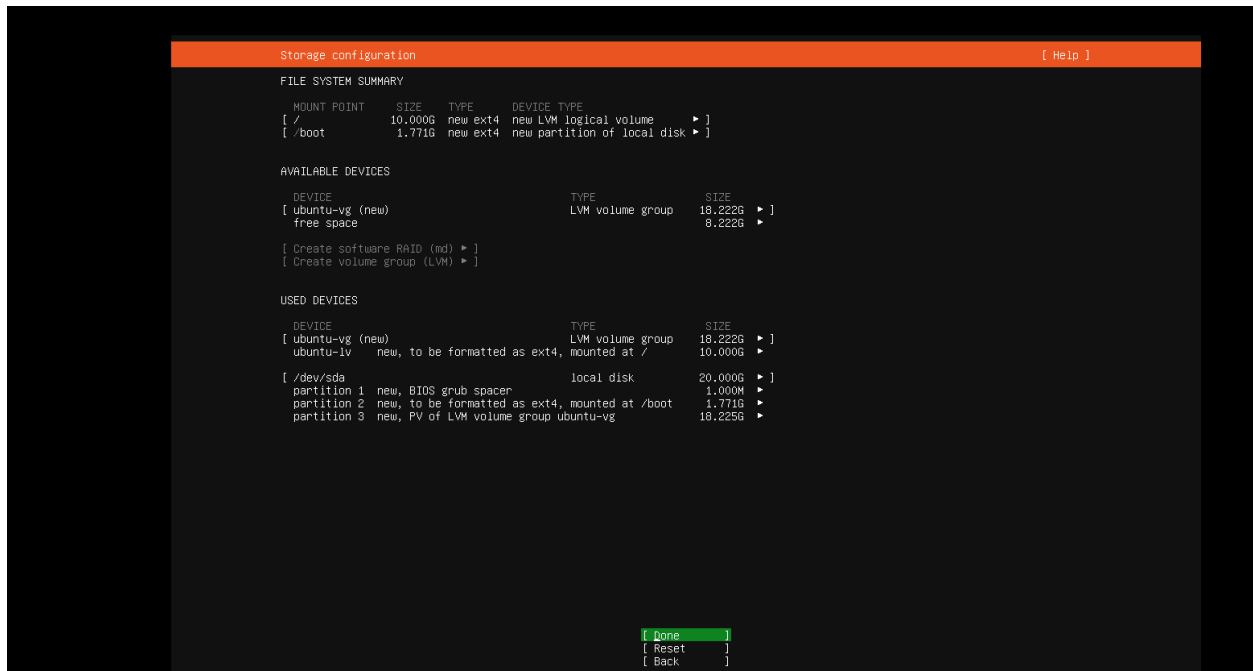
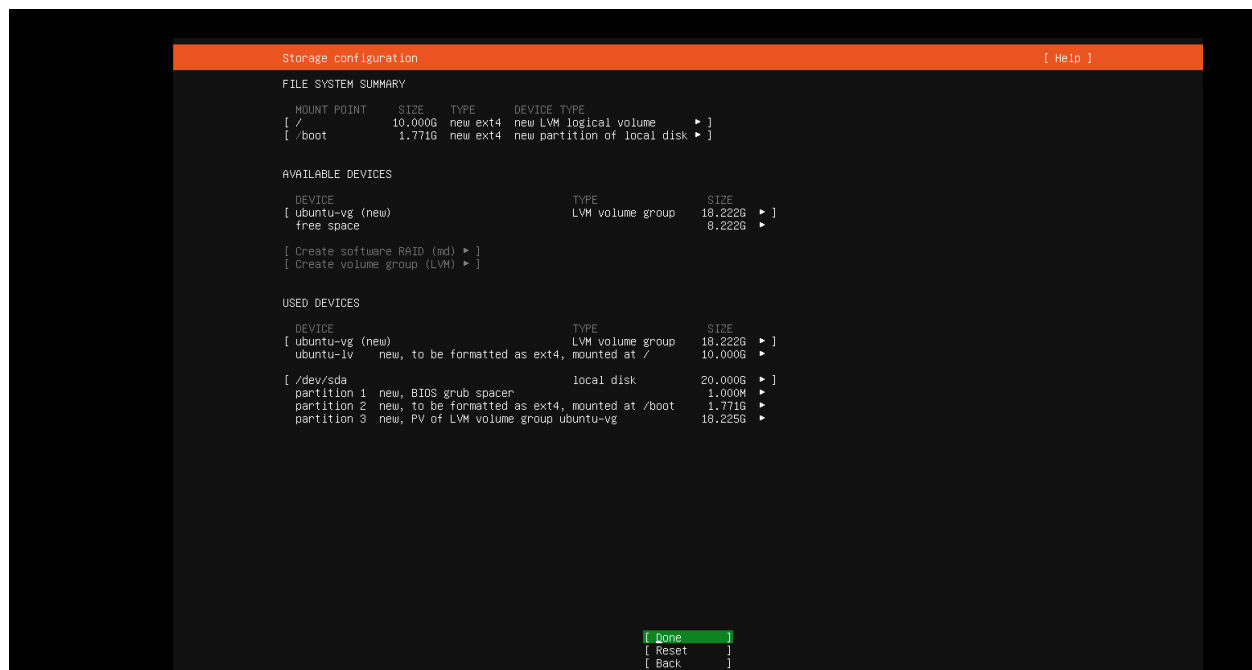


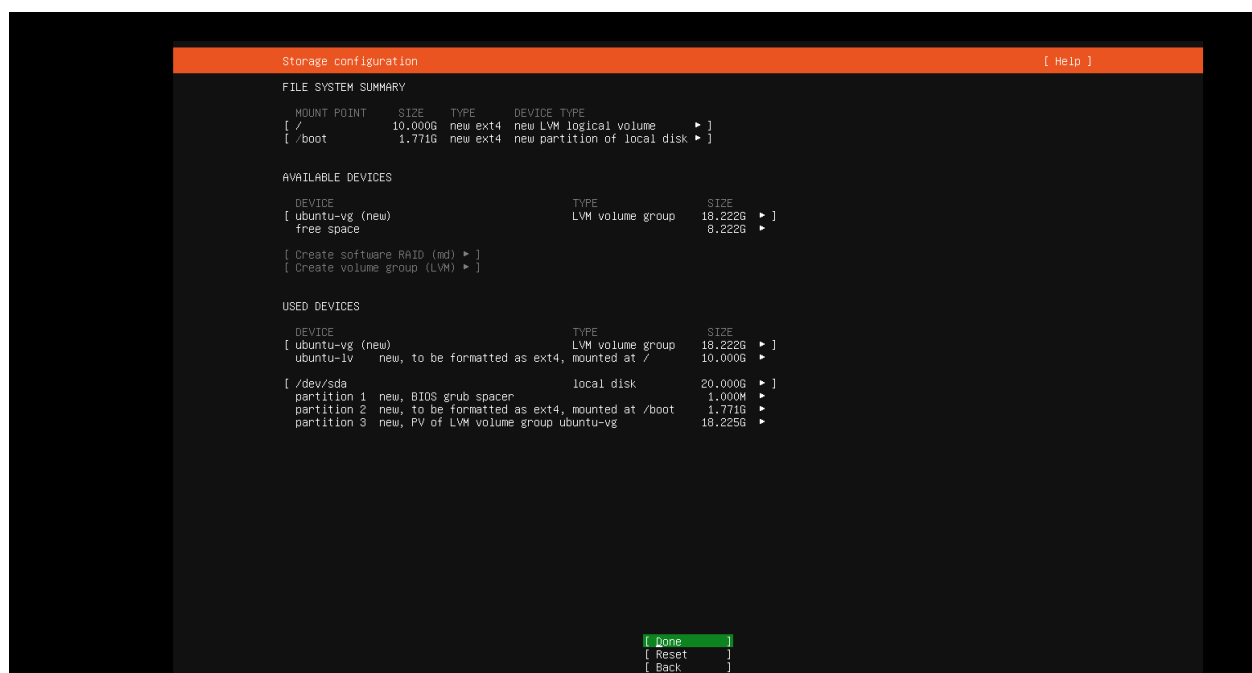
Figure 10.2: ubuntu_disk_details.png



Task 11: Review the Partitions

Description: Reviewed the proposed file system partition layout to ensure changes applied strictly to the selected virtual disk.

Figure 11.1: ubuntu_partition_layout.png



Task 12: Confirm the Storage Changes

Description: Read the destructive-action prompt and confirmed the storage changes to begin writing partitions to the virtual disk.

Figure 12.1: ubuntu_storage_warning.png

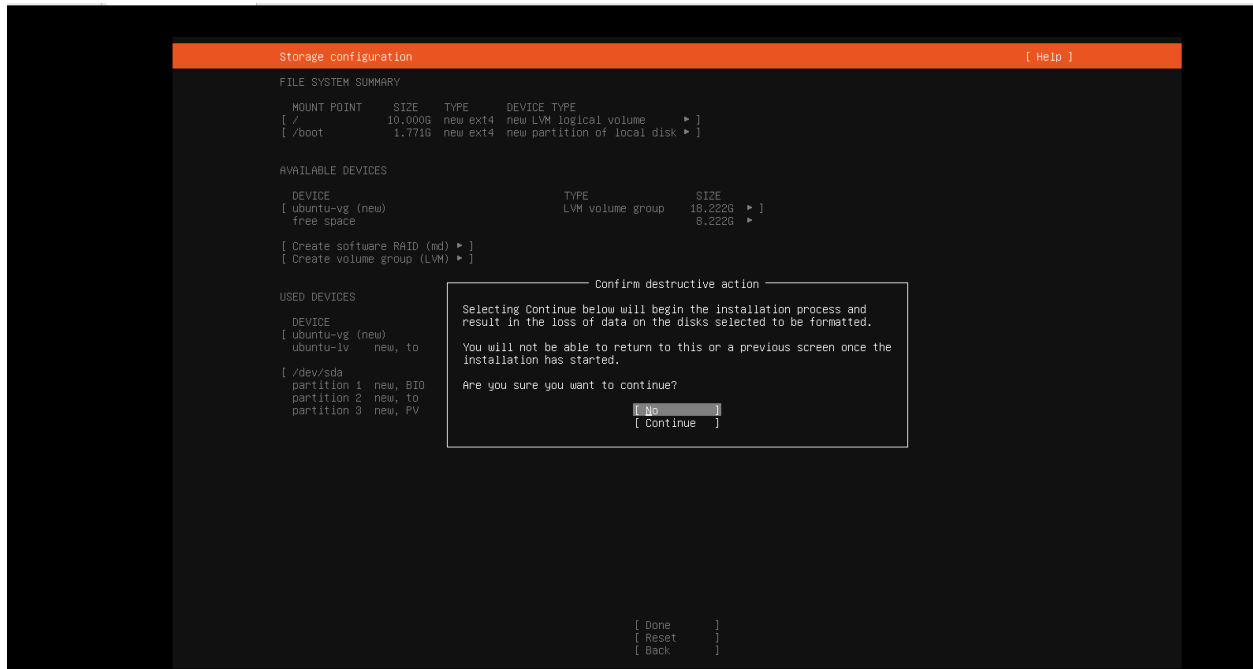
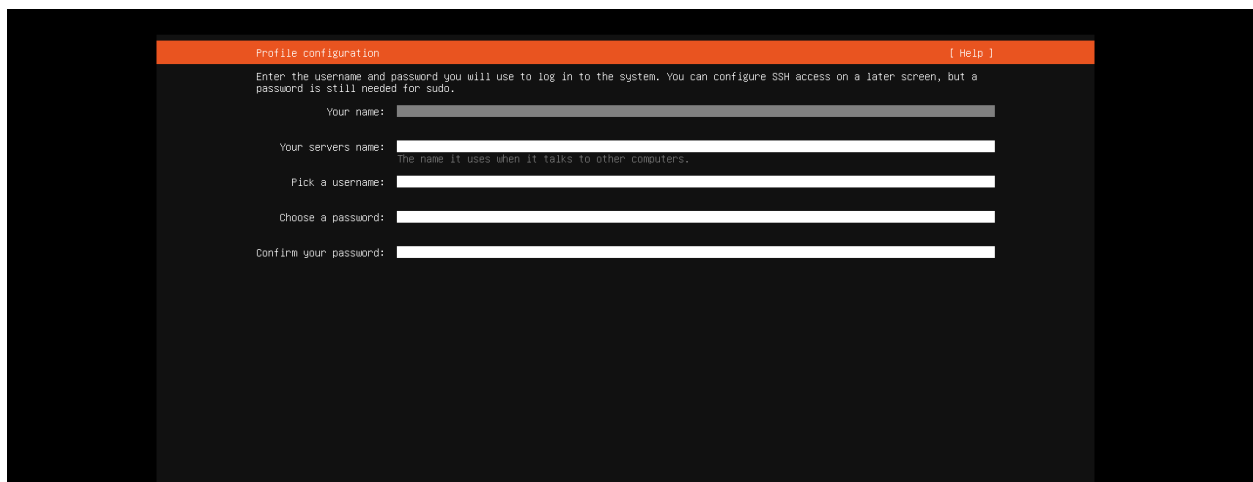


Figure 12.2: ubuntu_storage_confirmed.png



Task 13: Configure the Ubuntu Profile

Description: Entered **Ayesha Asif** in the "Your name" field, set server name to ubuntu, set the system username to ayeshaasif46, and configured a secure password.

Figure 13.1: ubuntu_actual_name.png

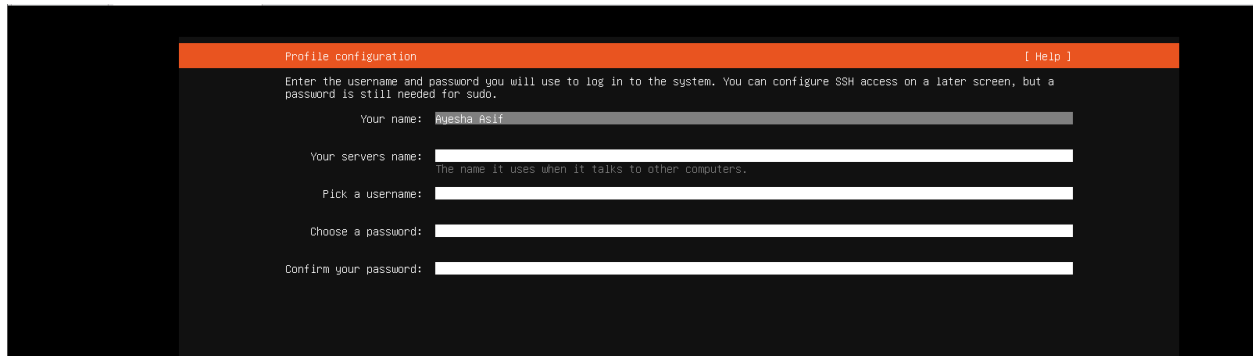


Figure 13.2: ubuntu_server_name.png

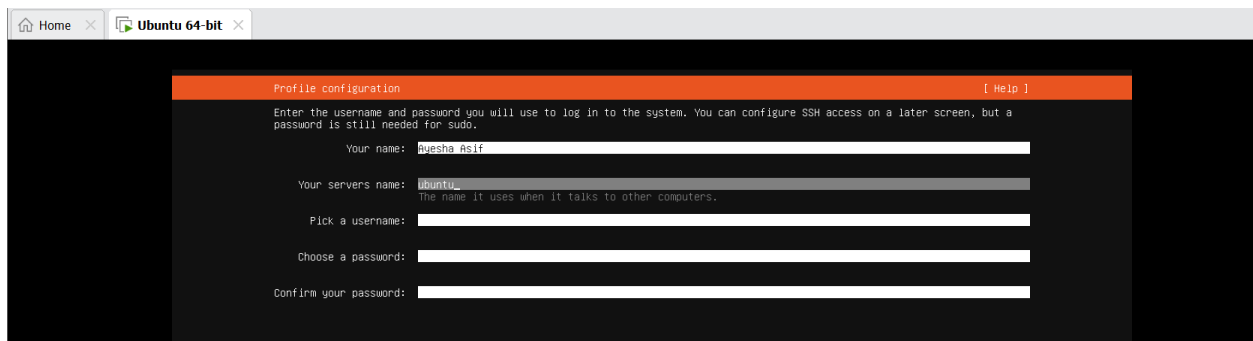


Figure 13.3: ubuntu_username.png

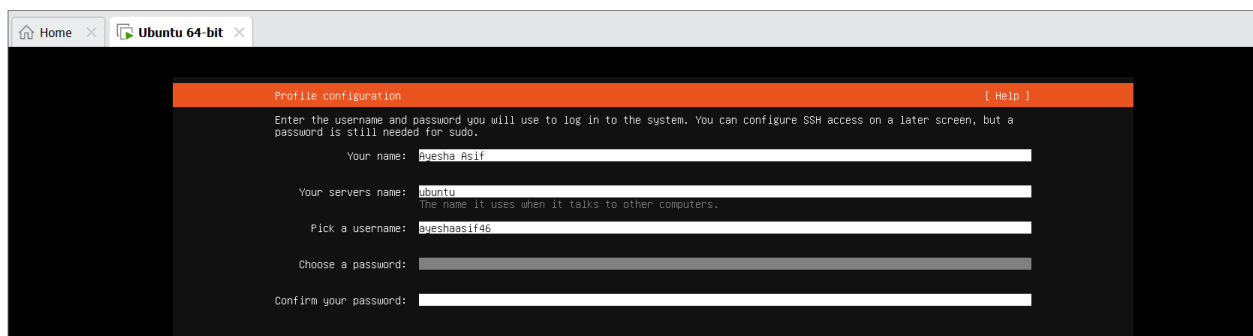
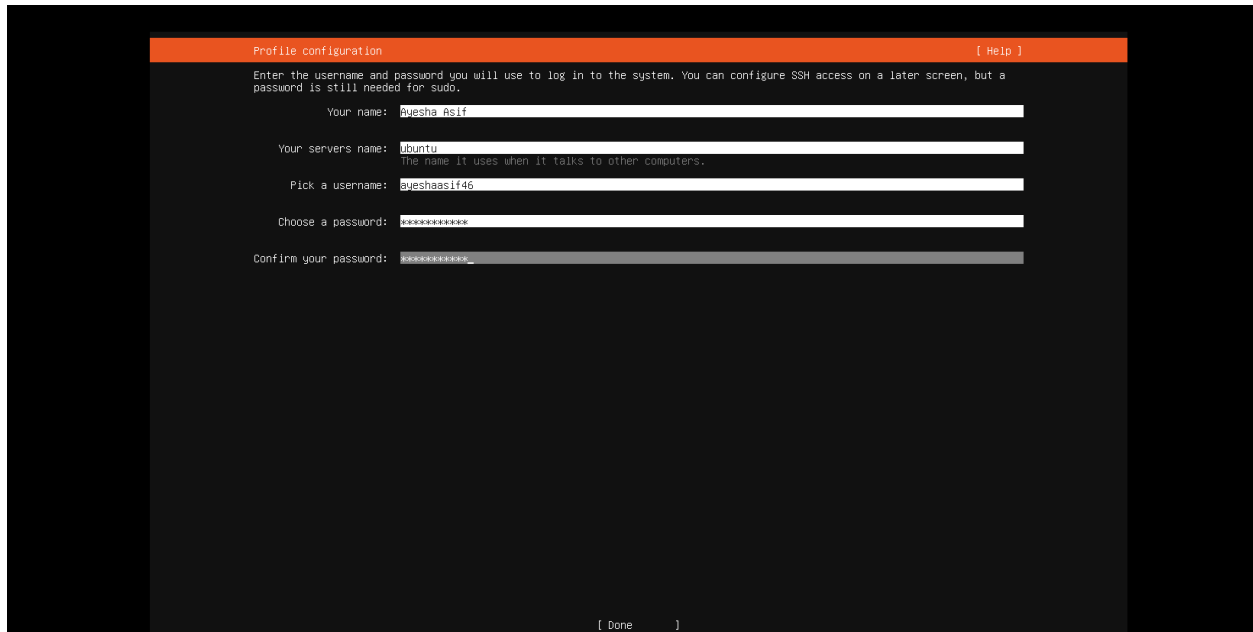


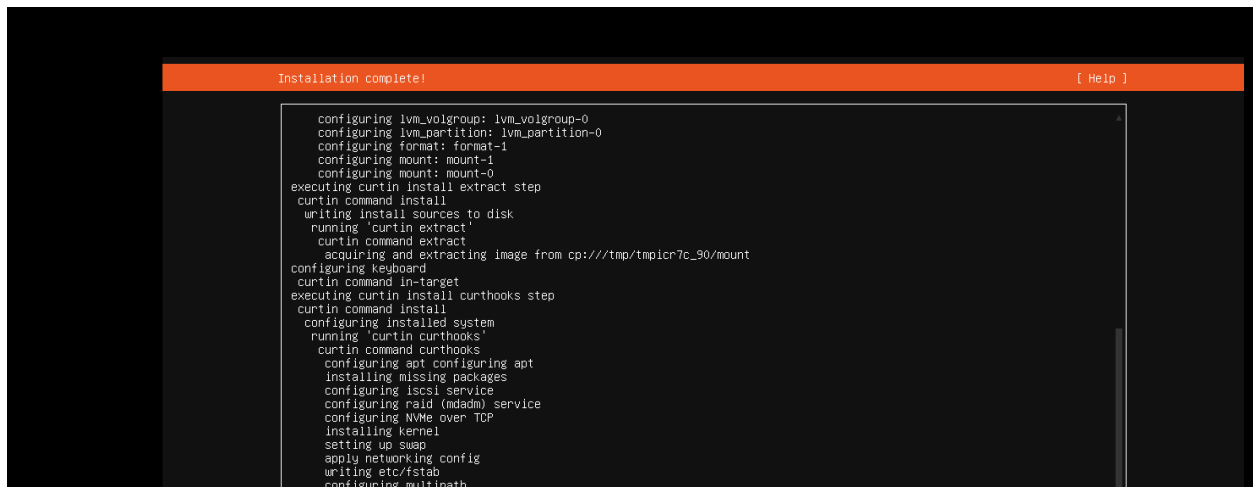
Figure 13.4: ubuntu_profile_setup.png



Task 14: Complete the Software Installation

Description: Monitored the installation progress as the server installed basic software packages and dependencies without interrupting the process.

Figure 14.1: ubuntu_installation_progress.png



Task 15: Reboot and Verify the Ubuntu Identity

Description: Rebooted the VM, logged in with credentials, and confirmed the active terminal prompt displayed ayeshaasif46@ubuntu:~\$.

Task 15.1: ubuntu_installation_complete.png

```
Installation complete! [ Help ]

configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpicr7c_90/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up snmp
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
restoring apt configuration
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Task 15.2: ubuntu_login_screen.png

```
[ 143.688439] cloud-init[1153]: The key fingerprint is:
[ 143.688585] cloud-init[1153]: SHA256:8h1JdyFr1rVQa2eYp+rXUrIgxPTK7vBR8MdhvF222M root@yubuntu
[ 143.688786] cloud-init[1153]: The key's randomart image is:
[ 143.688947] cloud-init[1153]: ---[RSA 3072]---
[ 143.689135] cloud-init[1153]:      . = * o = .
[ 143.689194] cloud-init[1153]:      . o = o +Eo
[ 143.689249] cloud-init[1153]:      . = o = + .
[ 143.689324] cloud-init[1153]:      . o + . = .
[ 143.689380] cloud-init[1153]:      . o = S o
[ 143.689445] cloud-init[1153]:      . +.oo+ .
[ 143.689490] cloud-init[1153]:      . . o = o .
[ 143.689539] cloud-init[1153]:      . o . = o
[ 143.689582] cloud-init[1153]:      . . ooo
[ 143.689627] cloud-init[1153]: -----[SHA256]-----
[ 143.689681] cloud-init[1153]: Generating public/private ecdsa key pair.
[ 143.689730] cloud-init[1153]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 143.689790] cloud-init[1153]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 143.689836] cloud-init[1153]: The key fingerprint is:
[ 143.689879] cloud-init[1153]: SHA256:p102x0p5Hp0EALh1jdIdMEvfJl80RmfyeYf60x60 root@yubuntu
[ 143.689915] cloud-init[1153]: The key's randomart image is:
[ 143.689959] cloud-init[1153]: ----[ECDSA 256]----
[ 143.694118] cloud-init[1153]:      . . . = . o .
[ 143.696611] cloud-init[1153]:      . o * o * .
[ 143.701747] cloud-init[1153]:      . = + + . o
[ 143.702525] cloud-init[1153]:      o . o + * . o +
[ 143.706255] cloud-init[1153]:      . . . S B oo.o+
[ 143.708183] cloud-init[1153]:      . B o . . . o
[ 143.710222] cloud-init[1153]:      = + . . .
[ 143.713156] cloud-init[1153]:      = oo .
[ 143.713632] cloud-init[1153]:      . o . E .
[ 143.714599] cloud-init[1153]: -----[SHA256]-----
[ 143.717185] cloud-init[1153]: Generating public/private ed25519 key pair.
[ 143.718496] cloud-init[1153]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 143.720317] cloud-init[1153]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 143.723240] cloud-init[1153]: The key fingerprint is:
[ 143.725389] cloud-init[1153]: SHA256:1ogmrMbELu50ShKmdnscVVC/Ms/MKY157eS+GHxJuMs root@yubuntu
[ 143.725835] cloud-init[1153]: The key's randomart image is:
[ 143.726190] cloud-init[1153]: ---[Ed25519 256]---
[ 143.726402] cloud-init[1153]:      . o .
[ 143.726642] cloud-init[1153]:      . .
[ 143.726871] cloud-init[1153]:      . .
[ 143.729153] cloud-init[1153]:      . . o o * o
[ 143.729147] cloud-init[1153]:      . . o o * So * + +
[ 143.732457] cloud-init[1153]:      o = + . B * * .
[ 143.735255] cloud-init[1153]:      ++o . . = + =
[ 143.742517] cloud-init[1153]:      +o+ . o . *
[ 143.743403] cloud-init[1153]:      . + . . E o .
[ 143.744191] cloud-init[1153]: -----[SHA256]-----

ubuntu login:
```

Task 15.3: ubuntu_terminal_login.png

```
Ubuntu 26.04.1 LTS ybuntu tty1
ybuntu login: ayeshaasif46
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu 17 Sept 07:33:56 UTC 2026

System load: 0.14           Memory usage: 20%    Processes:    248
Usage of /:  47.2% of 9.75GB Swap usage:   0%      Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Task 15.4: ubuntu_identity_verified.png

```
Ubuntu 26.04.1 LTS ybuntu tty1
ybuntu login: ayeshaasif46
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu 17 Sept 07:33:56 UTC 2026

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ayeshaasif46@ybuntu:~$
```

Task 16: Find the Ubuntu Server IP Address

Description: Executed the ip addr command in the system terminal and identified the assigned IPv4 address on the active network interface.

Figure 16.1: ubuntu_ip_addr_command.png

```

ayeshaasif46@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:67:ab:a8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2967aba8
    inet 192.168.64.100/24 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe67:aba8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaasif46@ubuntu:~$ _

```

Figure 16.2: ubuntu_ip_address.png

```

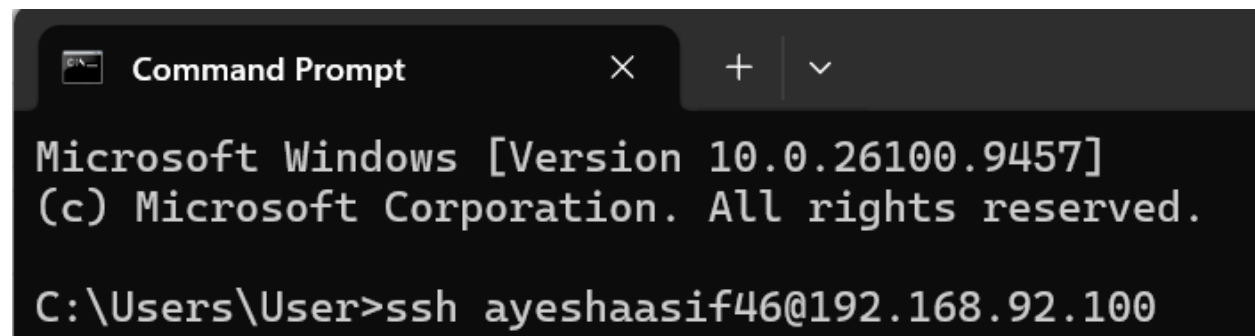
ayeshaasif46@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:67:ab:a8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2967aba8
    inet 192.168.64.100/24 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe67:aba8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ayeshaasif46@ubuntu:~$ _

```

Task 17: Connect to Ubuntu Server from Windows

Description: Opened Windows Command Prompt and established a remote SSH session using `ssh ayeshaasif46@<ubuntu-server-ip>`.

Figure 17.1: windows_ssh_command.png



```

Microsoft Windows [Version 10.0.26100.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh ayeshaasif46@192.168.92.100

```

Figure 17.2: windows_ssh_fingerprint.png

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.92.100' (ED25519) to the list of known hosts.
ayeshaasif46@192.168.92.100's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu 17 Sept 09:04:43 UTC 2026

System load:  0.14           Processes:            226
Usage of /:   48.7% of 9.75GB Users logged in:          0
Memory usage: 25%           IPv4 address for ens33: 192.168.64.100
Swap usage:   0%            IPv4 address for ens33: 192.168.92.100

Expanded Security Maintenance for Applications is not enabled.

109 updates can be applied immediately.
96 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Figure 17.3: windows_ssh_login.png

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.92.100' (ED25519) to the list of known hosts.
ayeshaasif46@192.168.92.100's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu 17 Sept 09:04:43 UTC 2026

System load:  0.14           Processes:            226
Usage of /:   48.7% of 9.75GB Users logged in:          0
Memory usage: 25%           IPv4 address for ens33: 192.168.64.100
Swap usage:   0%            IPv4 address for ens33: 192.168.92.100

Expanded Security Maintenance for Applications is not enabled.
```

Figure 17.4: windows_ssh_idetit.png

```
ayeshaasif46@yuntu: ~  
IPv4 Address. . . . . : 10.11.0.94  
Subnet Mask . . . . . : 255.255.0.0  
Default Gateway . . . . . : 10.11.0.1  
  
Ethernet adapter VMware Network Adapter VMnet8:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%20  
    IPv4 Address. . . . . : 192.168.92.1  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
  
Wireless LAN adapter Wi-Fi:  
  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . : itc.local  
  
C:\Users\User>ssh ayeshaasif46@192.168.92.100  
ssh: connect to host 192.168.92.100 port 22: Connection refused  
  
C:\Users\User>ssh ayeshaasif46@192.168.92.100  
The authenticity of host '192.168.92.100 (192.168.92.100)' can't be established.  
ED25519 key fingerprint is SHA256:logmrMbELu50ShKmdnscVVC/Ms/MkY157eS+GHxJuMs.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.92.100' (ED25519) to the list of known hosts.  
ayeshaasif46@192.168.92.100's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)  
  
 * Documentation:  https://docs.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Thu 17 Sept 09:04:43 UTC 2026  
  
System load:  0.14      Processes:      226  
Usage of /:   48.7% of 9.75GB    Users logged in: 0  
Memory usage: 25%      IPv4 address for ens33: 192.168.64.100  
Swap usage:   0%        IPv4 address for ens33: 192.168.92.100  
  
Expanded Security Maintenance for Applications is not enabled.  
  
109 updates can be applied immediately.  
96 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
ayeshaasif46@yuntu:~$ |
```

Task 18: Prepare Lab1 Solution PDF

Description: Assembled a document with a title page containing student details, included all ordered screenshots with clear captions, and saved it as Lab1_Solution.pdf.

Figure 18.1: solution_title_page.png



CLOUD COMPUTING LAB

Lab Number 01

**Lab Title: GitHub Identity, Student Portal Registration, and
Ubuntu Server Installation**

Name: Ayesha Asif

Registration No: FA24B1-SE-046 |

Course Code: SE-665

Section: B

GitHub Username: ayeshaasif46

DEPARTMENT OF SOFTWARE ENGINEERING

SESSION: 2024-2028